

Verification of WSL Installation

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sindh>wsl --status
Default Distribution: Ubuntu
Default Version: 2
WSL1 is not supported with your current machine configuration.
Please enable the "Windows Subsystem for Linux" optional component to use WSL1.

C:\Users\sindh>
```

Verification for gcc version

```
sindhuja@SINDHUJA: ~
sindhuja@SINDHUJA:~$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

sindhuja@SINDHUJA:~$
```

Gcc Program

```
sindhuj@SINDHUJA: ~
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
        exit(1);
    }
    else {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```

sindhuja@SINDHUJA: ~
sindhuja@SINDHUJA:~$ nano process.c
sindhuja@SINDHUJA:~$ gcc process.c -o process
sindhuja@SINDHUJA:~$ ./process
Enter a Linux command: pwd

Parent Process
Parent PID : 1127
Child PID : 1128

Child Process
Child PID : 1128
Parent PID: 1127
/home/sindhuja

Child process completed.
sindhuja@SINDHUJA:~$ |

```

SIMPLE COMMANDS

Topic 2: Directory Management			
10	ls	List files and directories	ls
11	ls -l	Display detailed file information	ls -l
12	ls -a	Display all files including hidden files	ls -a
13	cd	Change the current directory	cd Documents
14	cd ..	Move to the parent directory	cd ..
15	mkdir	Create a new directory	mkdir OSLab
16	rmdir	Remove an empty directory	rmdir Test
17	find	Search for files and directories	find . -name "*.c"
18	locate	Quickly locate files	locate hello.c

```

sindhuja@SINDHUJA: /home
sindhuja@SINDHUJA:~$ ls
OSSP      fork_exec.c  process      skill_week3  task1.c  task4  task5.c
fork_exec  linux_commands  process.c    task1        task3    task5
sindhuja@SINDHUJA:~$ ls -l
total 140
drwxr-xr-x 11 sindhuja sindhuja 4096 Aug 1 08:12 OSSP
-rwxr-xr-x 1 sindhuja sindhuja 16088 Aug 1 12:26 fork_exec
-rw-r--r-- 1 sindhuja sindhuja 518 Aug 1 08:14 fork_exec.c
drwxr-xr-x 2 sindhuja sindhuja 4096 Aug 8 09:10 linux_commands
-rwxr-xr-x 1 sindhuja sindhuja 16440 Aug 3 06:11 process
-rw-r--r-- 1 sindhuja sindhuja 868 Aug 3 06:11 process.c
drwxr-xr-x 2 sindhuja sindhuja 4096 Aug 8 08:42 skill_week3
-rwxr-xr-x 1 sindhuja sindhuja 16432 Aug 3 04:41 task1
-rw-r--r-- 1 sindhuja sindhuja 992 Aug 3 05:08 task1.c
-rwxr-xr-x 1 sindhuja sindhuja 16272 Aug 8 08:16 task3
-rwxr-xr-x 1 sindhuja sindhuja 16448 Aug 8 08:23 task4
-rwxr-xr-x 1 sindhuja sindhuja 16424 Aug 8 08:28 task5
-rw-r--r-- 1 sindhuja sindhuja 2040 Aug 8 08:27 task5.c
sindhuja@SINDHUJA:~$ ls -a
.      .cache      .sudo_as_admin_successful  process      task3
..     .config     OSSP                        process.c    task4
.bash_history  .local      fork_exec                skill_week3  task5
.bash_logout  .motd_shown  fork_exec.c              task1        task5.c
.bashrc       .profile     linux_commands            task1.c
sindhuja@SINDHUJA:~$ cd
sindhuja@SINDHUJA:~$ cd ..

```

```
sindhuj@SINDHUJA:/home$ cd ~/linux_commands
sindhuj@SINDHUJA:~/linux_commands$ mkdir OSLab
sindhuj@SINDHUJA:~/linux_commands$ ls
OSLab
sindhuj@SINDHUJA:~/linux_commands$ mkdir Test
sindhuj@SINDHUJA:~/linux_commands$ ls
OSLab Test
sindhuj@SINDHUJA:~/linux_commands$ touch program1.c
sindhuj@SINDHUJA:~/linux_commands$ ls
OSLab Test program1.c
sindhuj@SINDHUJA:~/linux_commands$ find . -name "*.c"
./program1.c
sindhuj@SINDHUJA:~/linux_commands$ |
```

```
sindhuj@SINDHUJA:/home$ find
.
./sindhuj
./sindhuj/linux_commands
./sindhuj/.local
./sindhuj/.local/share
./sindhuj/.local/share/nano
./sindhuj/fork_exec
./sindhuj/.cache
./sindhuj/.cache/motd.legal-displayed
./sindhuj/.config
./sindhuj/.config/procps
./sindhuj/task4
./sindhuj/task3
./sindhuj/.bash_logout
./sindhuj/.motd_shown
./sindhuj/.profile
./sindhuj/task5
./sindhuj/process.c
./sindhuj/fork_exec.c
./sindhuj/skill_week3
./sindhuj/skill_week3/task3.c
./sindhuj/skill_week3/File1.txt
./sindhuj/skill_week3/task4.c
./sindhuj/skill_week3/task2.c
./sindhuj/skill_week3/task5
./sindhuj/skill_week3/task1.c
./sindhuj/skill_week3/task5.c
./sindhuj/skill_week3/task2
./sindhuj/skill_week3/File2.txt
./sindhuj/skill_week3/task1
./sindhuj/.sudo_as_admin_successful
./sindhuj/task1.c
./sindhuj/.bash_history
./sindhuj/task5.c
./sindhuj/.bashrc
./sindhuj/process
./sindhuj/OSSP
./sindhuj/OSSP/.gitignore
./sindhuj/OSSP/include
./sindhuj/OSSP/include/parser.h
./sindhuj/OSSP/include/executor.h
./sindhuj/OSSP/include/shell.h
./sindhuj/OSSP/include/builtin.h
./sindhuj/OSSP/.git
./sindhuj/OSSP/.git/branches
./sindhuj/OSSP/.git/hooks
./sindhuj/OSSP/.git/hooks/pre-merge-commit.sample
./sindhuj/OSSP/.git/hooks/pre-rebase.sample
./sindhuj/OSSP/.git/hooks/pre-push.sample
./sindhuj/OSSP/.git/hooks/update.sample
./sindhuj/OSSP/.git/hooks/commit-msg.sample
./sindhuj/OSSP/.git/hooks/pre-applypatch.sample
./sindhuj/OSSP/.git/hooks/post-update.sample
./sindhuj/OSSP/.git/hooks/fsmonitor-watchman.sample
./sindhuj/OSSP/.git/hooks/pre-commit.sample
./sindhuj/OSSP/.git/hooks/sendemail-validate.sample
./sindhuj/OSSP/.git/hooks/push-to-checkout.sample
./sindhuj/OSSP/.git/hooks/pre-receive.sample
```

Topic 3: File Management			
19	touch	Create a new empty file	touch file1.txt
20	cat	Display file contents	cat file1.txt
21	cp	Copy files or directories	cp a.txt b.txt
22	mv	Move or rename files	mv old.txt new.txt
23	rm	Delete a file	rm test.txt
24	head	Display first lines of a file	head hello.c
25	tail	Display last lines of a file	tail hello.c
26	more	View a file page by page	more file.txt
27	less	Scroll through a file	less hello.c
28	wc	Count lines, words and characters	wc hello.c
29	diff	Compare two files	diff a.txt b.txt
30	sort	Sort file contents	sort data.txt
31	uniq	Remove duplicate lines	uniq data.txt
32	grep	Search for text inside a file	grep main hello.c

```

sindhuja@SINDHUJA: ~/linux  X + v
sindhuja@SINDHUJA:~/linux_commands$ touch file1.txt
sindhuja@SINDHUJA:~/linux_commands$ ls
OSLab Test file1.txt program1.c
sindhuja@SINDHUJA:~/linux_commands$ cat file1.txt
sindhuja@SINDHUJA:~/linux_commands$ ls
OSLab Test file1.txt program1.c
sindhuja@SINDHUJA:~/linux_commands$ nano file1.txt
sindhuja@SINDHUJA:~/linux_commands$ cat file1.txt
Hello world
Welcome to Opertainng Systems
This is my first File

sindhuja@SINDHUJA:~/linux_commands$ touch file.txt
sindhuja@SINDHUJA:~/linux_commands$ ls
OSLab Test file.txt file1.txt program1.c
sindhuja@SINDHUJA:~/linux_commands$ nano file1.txt
sindhuja@SINDHUJA:~/linux_commands$ cat file1.txt
Hello world
Welcome to Opertainng Systems
This is my first File

sindhuja@SINDHUJA:~/linux_commands$ cp file1.txt file2.txt
sindhuja@SINDHUJA:~/linux_commands$ ls
OSLab Test file.txt file1.txt file2.txt program1.c
sindhuja@SINDHUJA:~/linux_commands$ cat file2.txt
Hello world
Welcome to Opertainng Systems
This is my first File

sindhuja@SINDHUJA:~/linux_commands$ mv file2.txt renamed.txt
sindhuja@SINDHUJA:~/linux_commands$ ls
OSLab Test file.txt file1.txt program1.c renamed.txt
sindhuja@SINDHUJA:~/linux_commands$ cat renamed.txt
Hello world
Welcome to Opertainng Systems
This is my first File

sindhuja@SINDHUJA:~/linux_commands$ rm renamed.txt
sindhuja@SINDHUJA:~/linux_commands$ ls
OSLab Test file.txt file1.txt program1.c
sindhuja@SINDHUJA:~/linux_commands$ |

```

```
sindhuja@SINDHUJA:~/linux_commands$ nano file1.txt
sindhuja@SINDHUJA:~/linux_commands$ head file1.txt
Line 1 - Hello Linux
Line 2 - Operating System
Line 3 - File Management
Line 4 - Ubuntu
Line 5 - Terminal
Line 6 - Commands
Line 7 - Practice
Line 8 - Topic 3
Line 9 - head command
Line 10 - End of file
sindhuja@SINDHUJA:~/linux_commands$ tail file1.txt
Line 1 - Hello Linux
Line 2 - Operating System
Line 3 - File Management
Line 4 - Ubuntu
Line 5 - Terminal
Line 6 - Commands
Line 7 - Practice
Line 8 - Topic 3
Line 9 - head command
Line 10 - End of file
sindhuja@SINDHUJA:~/linux_commands$ tail -n 3 file1.txt
Line 8 - Topic 3
Line 9 - head command
Line 10 - End of file
sindhuja@SINDHUJA:~/linux_commands$ tail -n 5 file1.txt
Line 6 - Commands
Line 7 - Practice
Line 8 - Topic 3
Line 9 - head command
Line 10 - End of file
sindhuja@SINDHUJA:~/linux_commands$
```

```
Line 10 - End of file
sindhuja@SINDHUJA:~/linux_commands$ more file1.txt
Line 1 - Hello Linux
Line 2 - Operating System
Line 3 - File Management
Line 4 - Ubuntu
Line 5 - Terminal
Line 6 - Commands
Line 7 - Practice
Line 8 - Topic 3
Line 9 - head command
Line 10 - End of file
sindhuja@SINDHUJA:~/linux_commands$ seq 1 50 > numbers.txt
sindhuja@SINDHUJA:~/linux_commands$ more numbers.txt
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
```

```
sindhuja@SINDHUJA: ~/linux × + ▾
sindhuja@SINDHUJA:~$ cd ~/linux_commands
sindhuja@SINDHUJA:~/linux_commands$ nano fileA.txt
sindhuja@SINDHUJA:~/linux_commands$ nano fileB.txt
sindhuja@SINDHUJA:~/linux_commands$ diff fileA.txt fileB.txt
3,4c3
< Ubuntu
<
---
> Windows
sindhuja@SINDHUJA:~/linux_commands$ nano data.txt
sindhuja@SINDHUJA:~/linux_commands$ ^C
sindhuja@SINDHUJA:~/linux_commands$ cat data.txt
Zebra
Apple
Mango
Banana
Orange
sindhuja@SINDHUJA:~/linux_commands$ sort data.txt
Apple
Banana
Mango
Orange
Zebra
sindhuja@SINDHUJA:~/linux_commands$ nano data.txt
sindhuja@SINDHUJA:~/linux_commands$ cat data.txt
Apple
Banana
Apple
Mango
Banana
Orange
Orange
sindhuja@SINDHUJA:~/linux_commands$
```

```
sindhuja@SINDHUJA:~/linux_commands$ nano program.c
sindhuja@SINDHUJA:~/linux_commands$ cat program.c
#include <stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
sindhuja@SINDHUJA:~/linux_commands$ grep main program.c
int main()
sindhuja@SINDHUJA:~/linux_commands$ grep printf program.c
    printf("Hello Linux\n");
sindhuja@SINDHUJA:~/linux_commands$ |
```

Topic 4: File Editing

33	nano	Create or edit a file	nano hello.c
34	vi	Edit a file using Vim editor	vi hello.c

```
sindhuja@SINDHUJA: ~/linux × + v
sindhuja@SINDHUJA:~$ cd ~/linux_commands
sindhuja@SINDHUJA:~/linux_commands$ nano hello.c
sindhuja@SINDHUJA:~/linux_commands$ cat hello.c
#include <stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
sindhuja@SINDHUJA:~/linux_commands$ vi hello.c
sindhuja@SINDHUJA:~/linux_commands$ cat hello.c
#include <stdio.h>

int main()
{
    printf("Hello Linux from vi\n");
    return 0;
}
sindhuja@SINDHUJA:~/linux_commands$ |
```

Topic 5: C Program Compilation and Execution

35	gcc	Compile a C program	gcc hello.c -o hello
36	./program	Execute the compiled program	./hello
37	which	Locate the executable of a command	which gcc
38	whereis	Locate binary, source, and manual pages	whereis gcc
39	man	Display manual pages	man gcc

```
sindhuja@SINDHUJA:~/linux_commands$ gcc hello.c -o hello
sindhuja@SINDHUJA:~/linux_commands$ ls
OSLab Test data.txt file.txt file1.txt fileA.txt fileB.txt hello hello.c numbers.txt program.c program1.c
sindhuja@SINDHUJA:~/linux_commands$ ./hello
Hello Linux from vi
sindhuja@SINDHUJA:~/linux_commands$ which gcc
/usr/bin/gcc
sindhuja@SINDHUJA:~/linux_commands$ whereis gcc
gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gcc.1.gz
sindhuja@SINDHUJA:~/linux_commands$ man gcc
sindhuja@SINDHUJA:~/linux_commands$ |
```

```

sindhuja@SINDHUJA: ~/linux  ×  +  ▾  —
GCC(1) GNU
NAME
gcc - GNU project C and C++ compiler

SYNOPSIS
gcc [-c|-S|-E] [-std=standard]
    [-g] [-pg] [-Olevel]
    [-Wwarn...] [-Wpedantic]
    [-Idir...] [-ldir...]
    [-Dmacro[=defn]...] [-Umacro]
    [-foption...] [-mmachine-option...]
    [-o outfile] [@file] infile...

Only the most useful options are listed here; see below for the remainder.  g++ accepts mostly the same options as gcc.

DESCRIPTION
When you invoke GCC, it normally does preprocessing, compilation, assembly and linking.  The "overall options" allow you to stop this process at an intermediate stage.  For example, the -c option says not to run the linker.  Then the output consists of object files output by the assembler.

Other options are passed on to one or more stages of processing.  Some options control the preprocessor and others control the compiler itself.  Yet other options control the assembler and linker; most of these are not documented here, since they are rarely needed.

Most of the command-line options that you can use with GCC are useful for C programs; when an option is only useful for another language (usually C++), the explanation says so explicitly.  If the description for a particular option does not mention a source language, you can use that option with all supported languages.

The usual way to run GCC is to run the executable called gcc, or machine-gcc when cross-compiling, or machine-gcc-version to run a specific version of GCC.  When you compile C++ programs, you should invoke GCC as g++ instead.

The gcc program accepts options and file names as operands.  Many options have multi-letter names; therefore multiple single-letter options may not be grouped: -dv is very different from -d -v.

You can mix options and other arguments.  For the most part, the order you use doesn't matter.  Order does matter when you use several options of the same kind; for example, if you specify -L more than once, the directories are searched in the order specified.  Also, the placement of the -l option is significant.

Many options have long names starting with -f or with -W—for example, -fmove-loop-invariants, -Wformat and so on.  Most of these have both positive and negative forms; the negative form of -ffoo is -fno-foo.  This manual documents both forms, whichever one is not the default.

Some options take one or more arguments typically separated either by a space or by the equals sign (=) from the option name.  Unless documented otherwise, an argument can be either numeric or a string.  Numeric arguments must typically be small unsigned decimal or hexadecimal integers.  Hexadecimal arguments must begin with the 0x prefix.  Arguments to options that specify a size threshold of some sort may be arbitrarily large decimal or hexadecimal integers followed by a byte size suffix designating a multiple of bytes such as "kB" and "KiB" for kilobyte and kibibyte, respectively, and "MiB" for megabyte and mebibyte, "GB" and "GiB" for gigabyte and gibibyte, and so on.  Such arguments are designed by byte-size in the following text.  Refer to the NIST, IEC, and other relevant national and international standards for the full listing and explanation of the binary and decimal byte size prefixes.

OPTIONS
Option Summary
Here is a summary of all the options, grouped by type.  Explanations are in the following sections.

Overall Options
-c -S -E -o file -dumpbase dumpbase -dumpbase-ext auxdropsuf -dumpdir dumpdir -x language -v
--help[=class[,...]] --target-help --version -pass-exit-codes -pipe -specs=file -wrapper
-ffile-prefix-map=old=new -fcanon-prefix-map -fplugin=file -fplugin-arg=name=arg -fdump-ada-spec[=file]
-fada-spec-parent=unit -fdump-go-spec=file

C Language Options
-ansi -std=standard -aux-info filename -fno-asm -fno-builtin -fno-builtin-function -fcond-misc

```

Topic 6: Process Management			
40	ps	Display running processes	ps
41	ps -ef	Display all running processes	ps -ef
42	top	Monitor running processes	top
43	kill	Terminate a process	kill 1234
44	kill -9	Forcefully terminate a process	kill -9 1234

```
sindhuja@SINDHUJA: ~/linux
sindhuja@SINDHUJA:~/linux_commands$ ps
  PID TTY          TIME CMD
 1142 pts/0    00:00:00 bash
 1218 pts/0    00:00:00 ps
sindhuja@SINDHUJA:~/linux_commands$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root         1        0   0  0:14 ?        00:00:00 /sbin/init
root         2        1   0  0:14 hvc0    00:00:00 /init
root         6        2   0  0:14 hvc0    00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncat
root        52        1   0  0:14 ?        00:00:00 /usr/lib/systemd/systemd-journald
root       100        1   0  0:14 ?        00:00:01 /usr/lib/systemd/systemd-udevd
systemd+   159        1   0  0:14 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+   163        1   0  0:14 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
root       176        1   0  0:14 ?        00:00:00 /usr/sbin/cron -f -p
message+   177        1   0  0:14 ?        00:00:00 dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activat
root       185        1   0  0:14 ?        00:00:00 /usr/lib/systemd/systemd-logind
root       197        1   0  0:14 ?        00:00:00 /usr/libexec/wnsl-pro-service -vv
root       208        1   0  0:14 tty1    00:00:00 /sbin/agetty -o -p -- \u --noclear - linux
syslog     209        1   0  0:14 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root       224        1   0  0:14 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --w
root       300        2   0  0:14 pts/1   00:00:00 /bin/login -f
sindhuja   346        1   0  0:14 ?        00:00:00 /usr/lib/systemd/systemd --user
sindhuja   347       346   0  0:14 ?        00:00:00 (sd-pam)
sindhuja   376       300   0  0:14 pts/1   00:00:00 -bash
root     1137        2   0  0:13 ?        00:00:00 /init
root     1139     1137   0  0:13 ?        00:00:01 /init
sindhuja  1142     1139   0  0:13 pts/0    00:00:00 -bash
sindhuja  1252     1142   0  0:13 pts/0    00:00:00 ps -ef
sindhuja@SINDHUJA:~/linux_commands$ top
top - 11:57:24 up 1:42, 1 user, load average: 0.06, 0.05, 0.01
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.1 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7782.6 total, 7117.0 free, 520.1 used, 296.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7262.5 avail Mem

   PID USER      PR  NI  VIRT  RES  SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0 21528 13128 9796 S   0.0   0.2   0:00.96 systemd
    2 root        20   0  3180  2204 2072 S   0.0   0.0   0:00.02 init-systemd(Ub
   6 root        20   0  3196  2128 2008 S   0.0   0.0   0:00.00 init
  52 root       19  -1 66840 19800 18616 S   0.0   0.2   0:00.92 systemd-journal
 100 root        20   0 25316  6768  5168 S   0.0   0.1   0:01.24 systemd-udevd
 159 systemd+   20   0 21460 13216 10960 S   0.0   0.2   0:00.16 systemd-resolve
 163 systemd+   20   0 91028  7988  7016 S   0.0   0.1   0:00.55 systemd-timesyn
 176 root        20   0  4244  2680  2432 S   0.0   0.0   0:00.06 cron
 177 message+   20   0  9628  5380  4676 S   0.0   0.1   0:00.25 dbus-daemon
 185 root        20   0 17972  8836  7796 S   0.0   0.1   0:00.14 systemd-logind
 197 root        20   0 1755848 13716 10952 S   0.0   0.2   0:00.42 wsl-pro-service
 208 root        20   0  3124  1972  1836 S   0.0   0.0   0:00.00 atetty
 209 syslog     20   0 222516  6048  4644 S   0.0   0.1   0:00.18 rsyslogd
 224 root        20   0 107020 23108 13648 S   0.0   0.3   0:00.12 unattended-upgr
 300 root        20   0  6704  4732  3944 S   0.0   0.1   0:00.01 login
 346 sindhuja    20   0 20104 11448  9516 S   0.0   0.1   0:00.15 systemd
 347 sindhuja    20   0 21156  3596  1872 S   0.0   0.0   0:00.00 (sd-pam)
 376 sindhuja    20   0  6080  5336  3664 S   0.0   0.1   0:00.02 bash
1137 root        20   0  3184  1116  980 S   0.0   0.0   0:00.00 SessionLeader
1139 root        20   0  3200  1256  1108 S   0.0   0.0   0:01.12 Relay(1142)
1142 sindhuja    20   0  6076  5436  3708 S   0.0   0.1   0:00.06 bash
1253 sindhuja    20   0  9408  5736  3496 R   0.0   0.1   0:00.01 top
```

```
sindhuja@SINDHUJA:~/linux_commands$
sindhuja@SINDHUJA:~/linux_commands$ kill 1234
-bash: kill: (1234) - No such process
sindhuja@SINDHUJA:~/linux_commands$ sleep 300 &
[1] 1258
sindhuja@SINDHUJA:~/linux_commands$ ps
  PID TTY          TIME CMD
 1142 pts/0    00:00:00 bash
 1258 pts/0    00:00:00 sleep
 1259 pts/0    00:00:00 ps
sindhuja@SINDHUJA:~/linux_commands$ kill 5678
-bash: kill: (5678) - No such process
sindhuja@SINDHUJA:~/linux_commands$ ps
  PID TTY          TIME CMD
 1142 pts/0    00:00:00 bash
 1258 pts/0    00:00:00 sleep
 1260 pts/0    00:00:00 ps
sindhuja@SINDHUJA:~/linux_commands$ kill -9 1234
-bash: kill: (1234) - No such process
sindhuja@SINDHUJA:~/linux_commands$ sleep 300 &
[2] 1261
sindhuja@SINDHUJA:~/linux_commands$ kill -9 6789
-bash: kill: (6789) - No such process
sindhuja@SINDHUJA:~/linux_commands$ ps
  PID TTY          TIME CMD
 1142 pts/0    00:00:00 bash
 1258 pts/0    00:00:00 sleep
 1261 pts/0    00:00:00 sleep
 1262 pts/0    00:00:00 ps
sindhuja@SINDHUJA:~/linux_commands$ |
```

Topic 7: Job Control			
45	jobs	Display background jobs	jobs
46	bg	Resume a job in the background	bg %1
47	fg	Bring a job to the foreground	fg %1
48	&	Run a process in the background	./hello &

```
sindhuja@SINDHUJA:~/linux_commands$ sleep 300 &
[3] 1271
sindhuja@SINDHUJA:~/linux_commands$ sleep 300
^Z
[4]+  Stopped                  sleep 300
sindhuja@SINDHUJA:~/linux_commands$ bg %2
-bash: bg: job 2 already in background
sindhuja@SINDHUJA:~/linux_commands$ jobs
[1]  Running                  sleep 300 &
[2]  Running                  sleep 300 &
[3]-  Running                  sleep 300 &
[4]+  Stopped                  sleep 300
sindhuja@SINDHUJA:~/linux_commands$ ./hello &
[5] 1277
sindhuja@SINDHUJA:~/linux_commands$ Hello Linux from vi
jobs
[1]  Running                  sleep 300 &
[2]  Running                  sleep 300 &
[3]  Running                  sleep 300 &
[4]+  Stopped                  sleep 300
[5]-  Done                    ./hello
sindhuja@SINDHUJA:~/linux_commands$ sleep 30 &
[5] 1278
sindhuja@SINDHUJA:~/linux_commands$ jobs
[1]  Done                    sleep 300
[2]  Running                  sleep 300 &
[3]  Running                  sleep 300 &
[4]+  Stopped                  sleep 300
[5]-  Running                  sleep 30 &
sindhuja@SINDHUJA:~/linux_commands$ |
```

Topic 8: File Permissions and Links			
49	chmod	Change file permissions	chmod +x script.sh
50	chmod 755	Set read, write, and execute permissions	chmod 755 script.sh
51	ln	Create a hard link	ln file1 file2
52	ln -s	Create a symbolic link	ln -s file1 shortcut

```
sindhuja@SINDHUJA: ~/linux x + v
sindhuja@SINDHUJA:~$ cd ~/linux_commands
sindhuja@SINDHUJA:~/linux_commands$ nano script.sh
sindhuja@SINDHUJA:~/linux_commands$ ls -l script.sh
-rw-r--r-- 1 sindhuja sindhuja 37 Aug 10 12:05 script.sh
sindhuja@SINDHUJA:~/linux_commands$ chmod +x script.sh
sindhuja@SINDHUJA:~/linux_commands$ ls -l script.sh
-rwxr-xr-x 1 sindhuja sindhuja 37 Aug 10 12:05 script.sh
sindhuja@SINDHUJA:~/linux_commands$ chmod 755 script.sh
sindhuja@SINDHUJA:~/linux_commands$ ls -l script.sh
-rwxr-xr-x 1 sindhuja sindhuja 37 Aug 10 12:05 script.sh
sindhuja@SINDHUJA:~/linux_commands$ ln file1 file2
ln: failed to access 'file1': No such file or directory
sindhuja@SINDHUJA:~/linux_commands$ echo "This is my original file" > file1
sindhuja@SINDHUJA:~/linux_commands$ ln file1 file2
sindhuja@SINDHUJA:~/linux_commands$ ls -l file1 file2
-rw-r--r-- 2 sindhuja sindhuja 25 Aug 10 12:28 file1
-rw-r--r-- 2 sindhuja sindhuja 25 Aug 10 12:28 file2
sindhuja@SINDHUJA:~/linux_commands$ cat file1
This is my original file
sindhuja@SINDHUJA:~/linux_commands$ cat file2
This is my original file
sindhuja@SINDHUJA:~/linux_commands$ ls -li file1 file2
21114 -rw-r--r-- 2 sindhuja sindhuja 25 Aug 10 12:28 file1
21114 -rw-r--r-- 2 sindhuja sindhuja 25 Aug 10 12:28 file2
```

Topic 9: Disk and Memory Information

53	df -h	Display disk space usage	df -h
54	du -sh	Display directory size	du -sh .
55	free -h	Display memory usage (Linux)	free -h

```
sindhuj@SINDHUJA:~/linux_commands$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G   0    3.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.9G  4.0K   3.9G   1% /mnt/wsl
drivers         459G  142G   317G  31% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.2G   954G   1% /
none            3.9G   92K   3.9G   1% /mnt/wslg
none            3.9G   0    3.9G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M   3.8G   1% /init
none            3.9G  516K   3.8G   1% /run
none            3.9G   0    3.9G   0% /run/lock
none            3.9G   0    3.9G   0% /run/shm
none            3.9G   80K   3.9G   1% /mnt/wslg/versions.txt
none            3.9G   80K   3.9G   1% /mnt/wslg/doc
C:\             459G  142G   317G  31% /mnt/c
tmpfs           779M   20K   779M   1% /run/user/1000
sindhuj@SINDHUJA:~/linux_commands$ du -sh .
64K .
sindhuj@SINDHUJA:~/linux_commands$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.6Gi         518Mi        7.0Gi         3.5Mi        297Mi        7.1Gi
Swap:         2.0Gi           0B         2.0Gi
```

Topic 10: User and Environment Management

56	id	Display user and group IDs	id
57	env	Display environment variables	env
58	export	Set an environment variable	export PATH=\$PATH:.
59	alias	Create a command shortcut	alias ll='ls -l'

```
sindhuj@SINDHUJA:~/linux_commands$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G   0    3.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.9G  4.0K   3.9G   1% /mnt/wsl
drivers         459G  142G   317G  31% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.2G   954G   1% /
none            3.9G   92K   3.9G   1% /mnt/wslg
none            3.9G   0    3.9G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M   3.8G   1% /init
none            3.9G  516K   3.8G   1% /run
none            3.9G   0    3.9G   0% /run/lock
none            3.9G   0    3.9G   0% /run/shm
none            3.9G   80K   3.9G   1% /mnt/wslg/versions.txt
none            3.9G   80K   3.9G   1% /mnt/wslg/doc
C:\             459G  142G   317G  31% /mnt/c
tmpfs           779M   20K   779M   1% /run/user/1000
sindhuj@SINDHUJA:~/linux_commands$ du -sh .
64K .
sindhuj@SINDHUJA:~/linux_commands$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.6Gi         518Mi        7.0Gi         3.5Mi        297Mi        7.1Gi
Swap:         2.0Gi           0B         2.0Gi
sindhuj@SINDHUJA:~/linux_commands$ id
uid=1000(sindhuj) gid=1000(sindhuj) groups=1000(sindhuj),27(sudo),100(users)
```

```

sindhuja@SINDHUJA:~/linux_commands$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=SINDHUJA
PWD=/home/sindhuja/linux_commands
LOGNAME=sindhuja
HOME=/home/sindhuja
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/1322_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;
41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:
*.lha=01;31:*.lzh=01;31:*.lzm=01;21:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=0
1;31:*.diz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tztst=01;31:*.bz2=01;31:*.bz
=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31
:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=0
1;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=0
1;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:
*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.we
bm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01
;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xc
f=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;3
6:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.og
a=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~00;90:*.#00;90:*.bak=00;90:*.crdownload=00;90:*.dpgk-dist=00;90:*.
dpgk-new=00;90:*.dpgk-old=00;90:*.dpgk-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90
:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=sindhuja
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/
c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/
c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/Syst
em32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Fil
es/dotnet:/mnt/c/javaprograms:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL
/L/MySQL Shell 8.0/bin:/mnt/c/Users/sindh/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/sindh/AppD
ata/Local/Programs/Python/Python314:/mnt/c/Users/sindh/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/sindh/.loc
al/bin:/mnt/c/Users/sindh/AppData/Local/Programs/Eclipse Adoptium/jdk-25.0.1.8-hotspot/bin:/mnt/c/MinGW/bin:/mnt/c
/Users/sindh/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/sindh/AppData/Roaming/npm:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
OLDPWD=/home/sindhuja
=/usr/bin/env

```

```

sindhuja@SINDHUJA:~/linux_commands$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/P
rogram Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/
c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/Syst
em32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/do
tnet:/mnt/c/javaprograms:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MyS
QL Shell 8.0/bin:/mnt/c/Users/sindh/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/sindh/AppData/L
ocal/Programs/Python/Python314:/mnt/c/Users/sindh/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/sindh/.loc
al/bin:/mnt/c/Users/sindh/AppData/Local/Programs/Eclipse Adoptium/jdk-25.0.1.8-hotspot/bin:/mnt/c/MinGW/bin:/mnt/c
/Users/sindh/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/sindh/AppData/Roaming/npm:/snap/bin
sindhuja@SINDHUJA:~/linux_commands$ export PATH=$PATH:
sindhuja@SINDHUJA:~/linux_commands$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/P
rogram Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/
c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/Syst
em32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/do
tnet:/mnt/c/javaprograms:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MyS
QL Shell 8.0/bin:/mnt/c/Users/sindh/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/sindh/AppData/L
ocal/Programs/Python/Python314:/mnt/c/Users/sindh/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/sindh/.loc
al/bin:/mnt/c/Users/sindh/AppData/Local/Programs/Eclipse Adoptium/jdk-25.0.1.8-hotspot/bin:/mnt/c/MinGW/bin:/mnt/c
/Users/sindh/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/sindh/AppData/Roaming/npm:/snap/bin:
sindhuja@SINDHUJA:~/linux_commands$ alias ll='ls -l'
sindhuja@SINDHUJA:~/linux_commands$ ll
total 64
drwxr-xr-x 2 sindhuja sindhuja 4096 Aug 10 10:06 OSLab
drwxr-xr-x 2 sindhuja sindhuja 4096 Aug 10 10:07 Test
-rw-r--r-- 1 sindhuja sindhuja 46 Aug 10 10:53 data.txt
-rw-r--r-- 1 sindhuja sindhuja 0 Aug 10 10:14 file.txt
-rw-r--r-- 2 sindhuja sindhuja 25 Aug 10 12:28 file1
-rw-r--r-- 1 sindhuja sindhuja 203 Aug 10 10:40 file1.txt
-rw-r--r-- 2 sindhuja sindhuja 25 Aug 10 12:28 file2
-rw-r--r-- 1 sindhuja sindhuja 37 Aug 10 10:49 fileA.txt
-rw-r--r-- 1 sindhuja sindhuja 37 Aug 10 10:49 fileB.txt
-rwxr-xr-x 1 sindhuja sindhuja 15960 Aug 10 11:42 hello
-rw-r--r-- 1 sindhuja sindhuja 86 Aug 10 11:40 hello.c
-rw-r--r-- 1 sindhuja sindhuja 141 Aug 10 10:43 numbers.txt
-rw-r--r-- 1 sindhuja sindhuja 78 Aug 10 10:54 program.c
-rw-r--r-- 1 sindhuja sindhuja 0 Aug 10 10:08 program1.c
-rwxr-xr-x 1 sindhuja sindhuja 37 Aug 10 12:05 script.sh
sindhuja@SINDHUJA:~/linux_commands$ alias ll
alias ll='ls -l'
sindhuja@SINDHUJA:~/linux_commands$ unalias ll
sindhuja@SINDHUJA:~/linux_commands$ ll
ll: command not found
sindhuja@SINDHUJA:~/linux_commands$ |

```